Windowed Execution Fingerprints for Human/Agent Trace Analysis

Method, Data, and Exploratory Results

AttackAgent Detect research notes

2 November 2026

Abstract

We test whether a compact, interpretable matrix of command and coarse-action statistics can describe execution traces, distinguish released Human/Agent labels, and help retrieve same-task traces. The matrix contains eight measurements per variable-length command window. On 89 eligible AIT troubleshooting runs, its 24-value window summary reaches 56.3% task-macro balanced accuracy (BA), versus 55.2% for a whole-session eight-feature summary; the paired difference is only +1.1 percentage points (95% task-bootstrap interval $[-14.3, +15.4]$). Command TF–IDF reaches 96.8%, while the matrix falls to 39.9% under source-group holdout, exposing a strong collection/label dependence. In 17 Cyber Task Horizons (CTH) challenges, increasing the Agent share does not increase false merges under the fixed training-only threshold: command-cosine false-merge rate falls from 27.8% to 9.6%, while false-split rate rises from 64.7% to 98.4%. The matrix adds 0.35 pp to CTH Agent-only Hit@1 and 2.08 pp to Antieval Cybench Hit@1; both paired intervals include zero. These are exploratory, source-qualified benchmark results. Shared challenge identity is a task-retrieval proxy, not a shared operator, attack, or campaign label.

1 Question and scope

Some security association systems compare tool use, operation order, or system-behavior sequences. Agent-mediated execution can introduce other influences—model, framework, prompt, and tool configuration—alongside task and environment. This study tests whether a small log-derived fingerprint is measurable and whether it helps retrieval. It does not assume Agent participation must make traces more alike or less alike.

We separate three questions: (1) can the measurements predict the released Human/Agent labels in a held-out task? (2) how do a same-challenge retrieval baseline's errors vary as the Human/Agent composition of query–candidate roles changes? (3) does adding the fingerprint improve source-local Agent-only task retrieval? The first question uses AIT audit logs, while the latter two use CTF/tool-call datasets. Their labels and metrics are not pooled.

2 Data and relation labels

The registry contains 1,142 normalized records: 664 CTF and 478 supplemental task/security traces. Each analysis below uses only its qualified subset.

The archived AIT release contains 109 raw run logs; 95 parsed common-view runs were available in the prior pilot and 89 pass this matrix study's evidence gate. AIT tasks are Nextcloud/WordPress troubleshooting, not attacks. CTH Human records are terminal recordings and Agent records are structured

Table 1: Separate experimental tracks. Counts refer to eligible observations, not the full source releases.

Track	Data and eligible size	Label or relation	Intended inference
Mode recognition	AIT shell troubleshooting: 89 runs, 45 Human / 44 Agent, 12 task groups	Released execution-mode label	Predict labels on held-out task/source/application groups; the label remains entangled with collection and workflow
Mixture retrieval	CTH CTF: 17 challenge groups, 85 records in selected groups	Same source-qualified challenge vs. a different challenge	Compare FMR/FSR and ranking across controlled Human/Agent role mixtures
Agent-only retrieval	CTH: 121 traces / 48 tasks; Antieval Cybench: 31 / 12 tasks	Same source-qualified task within each source	Test whether matrix features change same-task Hit@1 within source

tool calls. Antieval Cybench contributes Agent-only HTTP/tool trajectories. These are different observation contracts. A shared CTF challenge identifies a common task context only; it is not ground truth for a common actor, Agent instance, attack activity, host, organization, or campaign.

3 Fingerprint method

We use the ordered executable/tool names exposed by each source adapter and map recognized commands to eight coarse action groups: discover, read, search, execute, build, network, modify, and environment. These are lossy observable traces; they do not expose an Agent's internal reasoning. We exclude IDs, task/source labels, model names, prompts, paths, timestamps, and capture-format strings from features.

For command window $w = (c_1, \ldots, c_n)$, let $p(c)$ be the empirical command-name distribution and let a_i be the mapped coarse action when available. The feature vector $\mathbf{k}_w = (K_1, \ldots, K_8)$ is:

$$K_1 = |\{c_i\}|/n, \qquad K_2 = -\sum_c p(c) \log_2 p(c), \qquad K_3 = \max_c p(c),$$
$$K_4 = \Pr(a_i \neq a_{i+1}), \qquad K_5 = H(A_{i+1} \mid A_i), \qquad K_6 = L_{\max}(a)/n,$$
$$K_7 = 1 - |\{(a_i, a_{i+1})\}|/|P_a|, \qquad K_8 = \Pr(c_i = c_{i+1}).$$

Here P_a contains adjacent pairs whose two action labels are known; $L_{\max}$ is the longest consecutive run of one mapped action. A statistic with insufficient mapped observations is missing, not zero.

Commands are partitioned into consecutive windows of eight. The matrix has a variable number of rows, $J \times 8$: at least 16 commands are required, at most the first 128 are used, and a final tail shorter than four commands is discarded. We do not impose a fixed number of attack phases. For the classifier, each column is summarized by its mean, population standard deviation, and least-squares slope over normalized window position, yielding 24 values. Missing values are imputed from training data only, then standardized and passed to class-balanced logistic regression ($C = 1$, threshold 0.5). The direct sequence matcher uses masked dynamic time warping (DTW) over train-scaled columns; a local cost requires at least three shared finite columns.

Table 2: The eight fingerprint dimensions. The definitions are operational measurements, not claims that a feature is unique to an Agent.

ID	Dimension	Window-level measurement
K1	Executable diversity	Distinct command names divided by command count
K2	Command entropy	Shannon entropy (bits) of command-name frequencies
K3	Dominant command share	Frequency of the most common command name
K4	Action switching	Fraction of mapped adjacent action pairs that change group
K5	Transition uncertainty	Conditional entropy of the next mapped action given the current action
K6	Longest action run	Longest consecutive run of one mapped action, divided by window length
K7	Repeated action-pair share	One minus the fraction of unique mapped adjacent action pairs
K8	Repeated command adjacency	Fraction of adjacent commands with identical names

4 Experimental protocol

4.1 Human/Agent label recognition

The primary AIT split holds out task groups; exact duplicate command sequences are purged across train and test. Secondary panels hold out source groups, application, or both task and source. We compare command-count length, command-name unigram/bigram TF–IDF, whole-session K1–K8, the 24-value window summary, and matrix-DTW 3-nearest-neighbor. Classifier hyperparameters are fixed; there is no score-based feature or window selection. We report task-macro BA and task-cluster bootstrap intervals over fixed out-of-fold predictions. These intervals do not include retraining uncertainty.

4.2 CTF mixture and pair-type errors

The CTH cohort has 17 tasks with at least two Human and two Agent sessions each. For each held-out query we rank five candidates: one same-task positive and four different-task negatives. The role slots are assigned to Human or Agent at fixed Agent shares of 0, 8/34, 17/34, 26/34, and 34/34 (reported as 0, 23.5, 50, 76.5, and 100%). There are 100 repeated draws per level; repeated draws are not independent samples. We compare command TF–IDF cosine and coarse action-bigram cosine. A threshold is selected using task-cross-fitted training pairs only: maximize positive recall subject to training negative FMR $\leq 10\%$. We separately report FMR, FSR, task BA, and Hit@1. Pair types HH, HA, and AA are calculated from the query and candidate execution labels, not as separate concentration experiments.

The matrix-mixture increment was not run: only one task has at least two Human and two Agent sessions with 16 or more commands, below the predeclared minimum of 15 such tasks. Therefore this experiment cannot test whether the matrix repairs Human/Agent mixture errors.

4.3 Agent-only task retrieval

For each source separately, tasks are split into three fixed folds before candidate formation. Each query has one same-source-task positive and either four (CTH) or two (Antieval) different-task candidates. We compare command cosine with a fixed logistic fusion of command cosine and negative masked-DTW distance, trained on training-task pairs. The primary metric is task-macro Hit@1; uncertainty is a task-cluster bootstrap. Cross-source direction tests train on one source and evaluate source-native retrieval in the other; no cross-source challenge identity is assumed. These transfer runs are post-observation exploratory because both sources were used for protocol development.

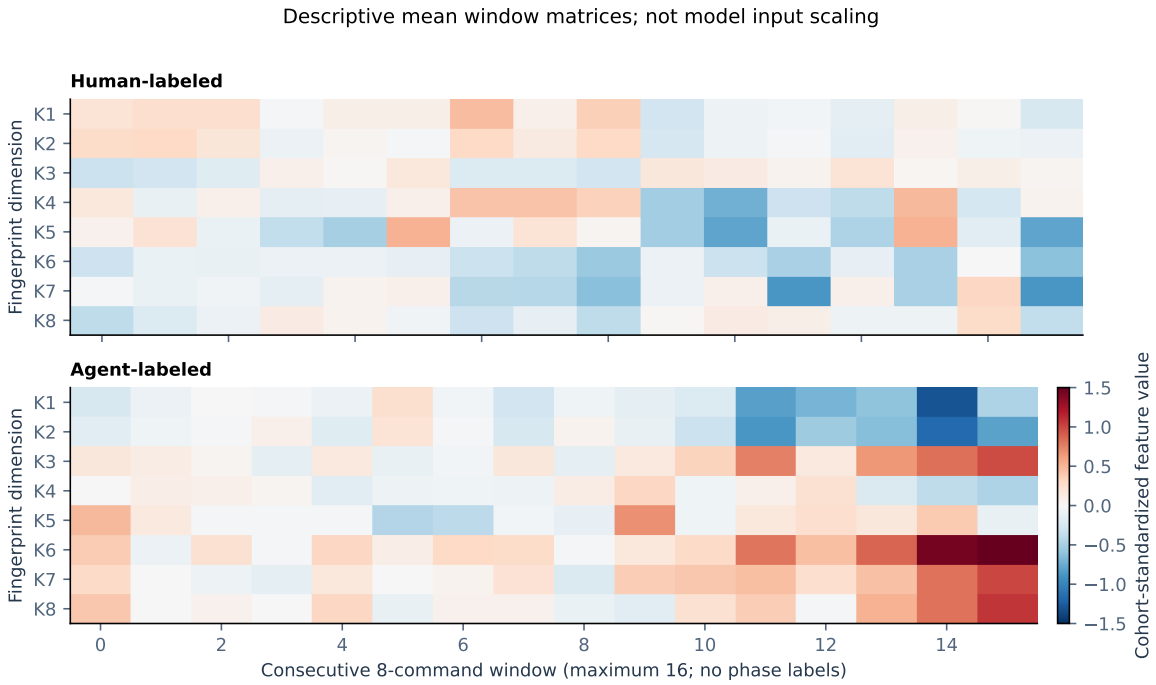


Figure 1: Observed AIT matrix example summarized by cohort for visualization. Each row is one eight-command window and each column one of K1–K8. Values are cohort-standardized for display; this figure is not a learned classifier and does not imply a universal Agent profile.

5 Results

5.1 The matrix is measurable, but label recognition is dominated by source-visible command vocabulary

On the primary held-out-task panel, command TF–IDF reaches 96.8% task-macro BA, while the matrix summary reaches 56.3% (95% CI [40.8,71.7]) and the whole-session K1–K8 baseline reaches 55.2% (95% CI [46.1,65.9]). The paired window-minus-whole-session difference is +1.1 pp (95% task-bootstrap interval $[-14.3, +15.4]$, 12 tasks). Matrix DTW 3-NN reaches 53.3%. The lexical result is a strong predictor of these released labels, but the Human/Agent collection pathway and operating workflow are not experimentally separated, so it is not an isolated actor-identity result.

Table 3: Primary AIT task-held-out results. The 95% intervals are task-cluster intervals on fixed out-of-fold predictions.

Model	Task-macro BA	Macro-F1	95% BA interval
Length only	46.6%	51.7%	[32.6,59.7]
Command TF–IDF	96.8%	95.5%	[94.2,99.2]
Whole-session K1–K8	55.2%	55.0%	[46.1,65.9]
Window summary	56.3%	57.3%	[40.8,71.7]
Matrix DTW 3-NN	53.3%	56.0%	[46.4,60.1]

The secondary holdouts show that the feature representation does not carry a stable execution-mode signal across these collection groups. Command TF–IDF remains predictive, but it too falls under stronger source/task controls. This pattern is consistent with vocabulary, workflow, and instrumentation carrying

Table 4: Task-macro BA across held-out panels. These secondary panels use the same frozen 89-run cohort.

Model	Task	Source group	Application	Task + source
Command TF–IDF	96.8%	82.6%	95.1%	80.0%
Window summary	56.3%	39.9%	45.8%	37.6%
Matrix DTW 3-NN	53.3%	36.9%	50.8%	37.8%

substantial information; it does not show that K1–K8 are an Agent-specific signature.

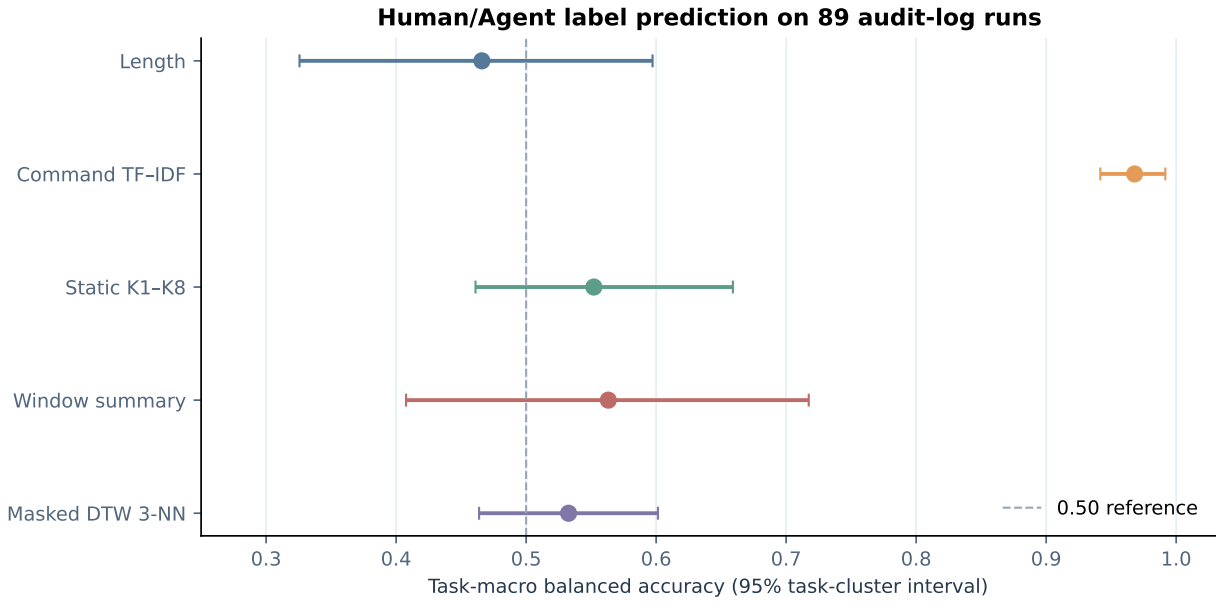


Figure 2: AIT held-out-task classifier comparison. Error bars are task-cluster bootstrap intervals on fixed predictions.

5.2 Agent mixture shifts the observed error from merging toward splitting

For command cosine, moving from 0% to 100% Agent roles changes FMR from 27.8% to 9.6%, while FSR changes from 64.7% to 98.4%. Action-bigram cosine shows the same direction (FMR 32.4% to 9.6%; FSR 52.9% to 98.4%). Thus these data do not support the preregistered directional story that increasing Agent concentration raises false merges. Instead, the fixed low-FMR training threshold increasingly rejects same-task pairs as the mixture changes. Hit@1 rankings and thresholded links are different measurements.

Table 5: CTH command-cosine results by actual Agent role share. FMR/FSR are thresholded link errors; Hit@1 is rank-based. Task-macro BA is $1 - (\text{FMR} + \text{FSR})/2$.

Agent share	FMR	FSR	Task BA	Hit@1
0%	27.8%	64.7%	53.8%	41.5%
23.5%	20.2%	77.0%	51.4%	36.5%
50%	11.8%	88.1%	50.1%	35.1%
76.5%	7.8%	96.4%	47.9%	34.9%
100%	9.6%	98.4%	46.0%	39.1%

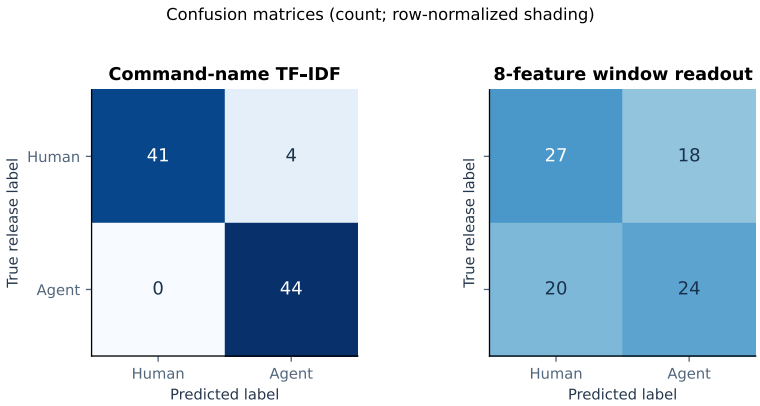


Figure 3: Fixed-threshold confusion matrices for command TF–IDF and the window-summary classifier on held-out tasks. Rows are released Human/Agent labels; columns are predicted labels.

At the 50% level, the command baseline's pair-type estimates are HH: FSR 64.7% / FMR 28.2%; HA: 93.6% / 5.9%; AA: 98.6% / 9.7%. The dominant observed issue is missed same-task retrieval for mixed-mode and Agent–Agent positives under this operating threshold. Since these are CTF task proxies and the capture interfaces differ, we cannot conclude that this is real attack-activity fragmentation.

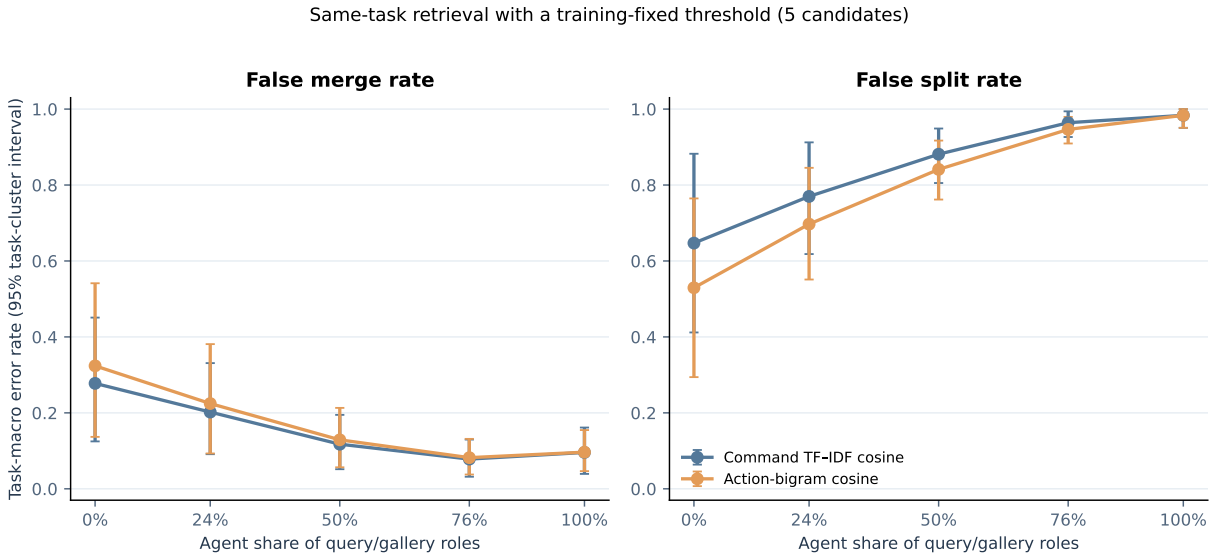


Figure 4: Thresholded false-merge and false-split rates across CTH role mixtures for command and action-bigram cosine. Points summarize repeated draws; intervals are across task clusters, not independent draws.

5.3 Adding the matrix does not reliably improve Agent-only retrieval

In CTH, command-cosine Hit@1 is 54.9% (five candidates; chance 20%) and matrix fusion is 55.2%, a paired gain of +0.35 pp (95% task-bootstrap interval $[-3.82, +4.17]$, 48 tasks). In Antieval Cybench, the corresponding values are 60.4% and 62.5% (three candidates; chance 33.3%), a +2.08 pp change (95% interval $[-6.25, +12.50]$, 12 tasks). These uncertain task-retrieval results do not validate the matrix as an association correction.

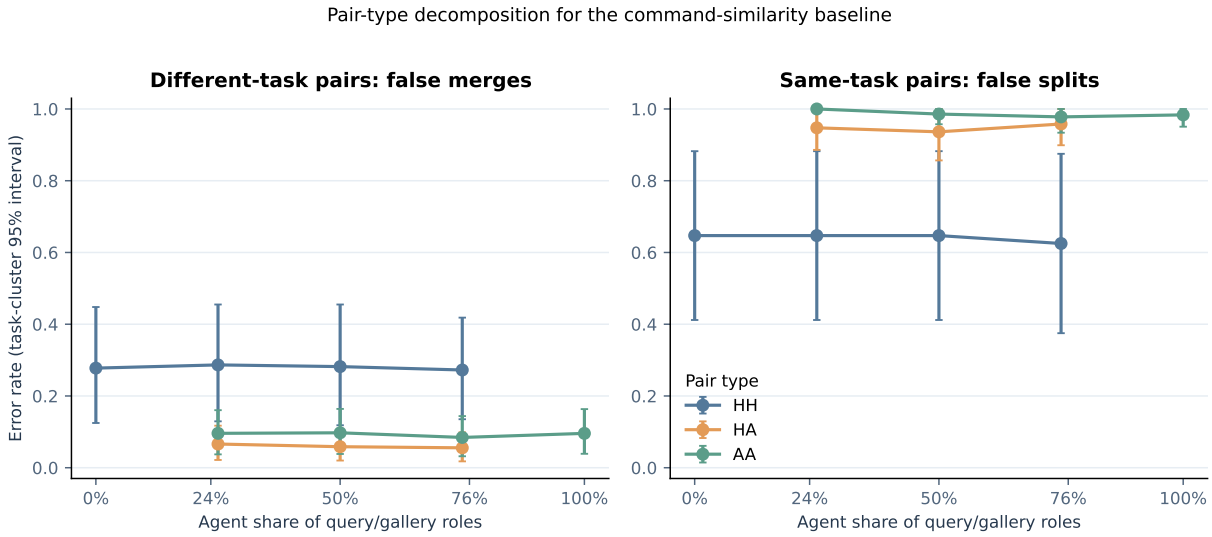


Figure 5: CTH error rates separated by HH, HA, and AA query–candidate pairs. A pair type is a composition of two execution labels, while Agent concentration is a cohort-level role allocation.

Table 6: Source-local Agent-only retrieval and paired change after matrix fusion. Candidate counts differ, so rows must not be pooled.

Source	Queries / tasks	Command Hit@1	Fused Hit@1	Paired change (95% CI)
CTH	121 / 48	54.9%	55.2%	+0.35 pp [−3.82, +4.17]
Antieval Cybench	31 / 12	60.4%	62.5%	+2.08 pp [−6.25, +12.50]

6 Relation to prior work

Trace studies command-sequence classification for underlying model-family inference [4], and ProcGrep studies procedural traces for coding-agent identification [5]; both motivate behavioral measurement but use different labels and domains. ATLAS and HOLMES use system-behavior evidence to reconstruct or associate attack stories [7, 8], but they do not identify Human versus Agent execution. Synthetic APT experiments motivate testing whether Agent-mediated behavior changes attribution cues [6]; they do not provide evidence for the outcomes in this study. Our CTF task-match evaluation is a deliberately narrower proxy, not a reproduction of campaign attribution. Source diversity is a design direction for broadly usable trace tools, but the present small transfer experiment does not demonstrate broad generalization.

7 Limitations and next measurement

The AIT Human/Agent label is coupled to model/person groups and their software-task workflow. CTH also couples label to terminal-versus-tool-call capture. The CTF positive label is same benchmark challenge, and the tested logs omit much internal Agent state. Small task counts produce wide uncertainty. Repeated mixture draws refine sampling estimates within these 17 tasks but do not create new independent tasks. Cross-source transfer is exploratory, and the matrix increment for mixed Human/Agent retrieval was blocked by the frozen evidence gate.

The next high-value dataset is a matched logging collection: the same command/action schema, task, permissions, and environment recorded under Human and Agent execution, with repeated runs and model/framework/prompt configuration retained as audit metadata but withheld from the classifier.

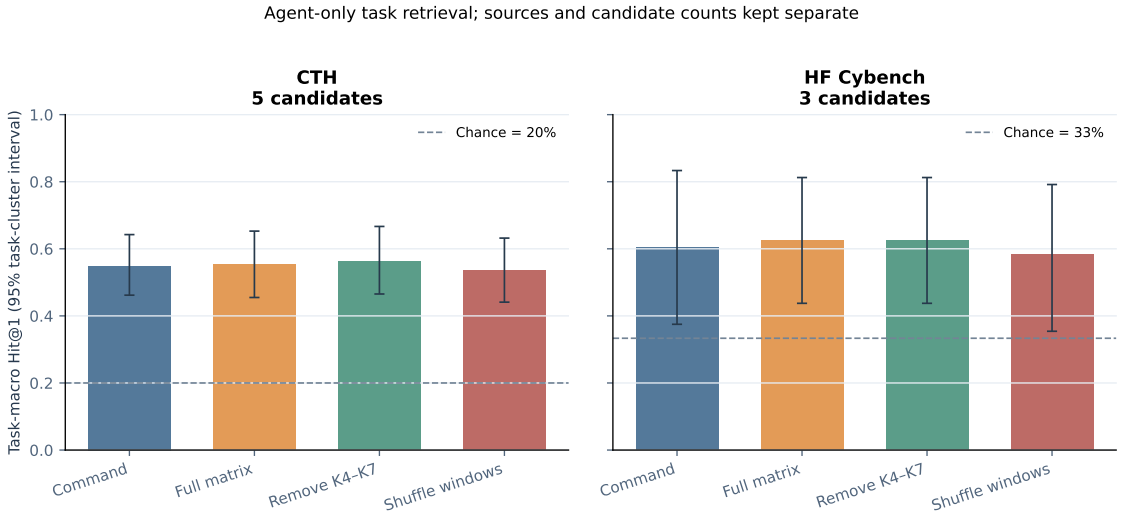


Figure 6: Paired task-macro Hit@1 differences for matrix fusion and feature-group ablations, kept separate by source. The candidate sets are source-specific.

Longer paired traces would make the matrix-mixture test eligible. This would distinguish execution-mode evidence from instrumentation and task vocabulary more directly.

A Complete mixture results

Table 7: CTH action-bigram cosine across Agent role mixtures.

Agent share	FMR	FSR	Task BA	Hit@1
0%	32.4%	52.9%	57.3%	38.9%
23.5%	22.5%	69.7%	53.9%	34.3%
50%	12.9%	84.1%	51.5%	32.1%
76.5%	8.2%	94.7%	48.6%	28.6%
100%	9.6%	98.4%	46.0%	26.3%

At the 50% mixture, command-cosine pair-type 95% task-cluster intervals are: HH FSR [41.2,88.2]%, FMR [11.8,45.5]%; HA FSR [85.7,100.0]%, FMR [2.0,10.3]%; AA FSR [95.7,100.0]%, FMR [3.8,16.4]%. These are descriptive intervals across the 17 task groups.

B Recognition ablations and controls

On the primary held-out-task predictions, removing K1–K3 changes task-macro BA by +5.0 pp (95% paired task-bootstrap interval $[0.0, +11.5]$); removing K4–K7 changes it by -5.5 pp $[-18.0, +7.7]$; removing K8 changes it by -2.3 pp $[-11.7, +4.0]$. The evidence does not isolate a stable winning feature group. Shuffling window order does not establish a robust ordered-sequence contribution. The fixed-first-32 comparison is restricted to 47 runs across seven tasks, so it is not directly comparable to the 89-run primary cohort.

C Cross-source retrieval: exploratory

Table 8: Train in one source; rank target-source-native same-task candidates in the other. No task identities are linked across sources.

Train	Test	Test tasks / queries	Command Hit@1	Fused Hit@1	Paired change (95% CI)
CTH	Antieval	12 / 31	51.4%	51.4%	0.00 pp [0.00,0.00]
Antieval	CTH	48 / 121	52.8%	54.5%	+1.74 pp [−3.82, +6.94]

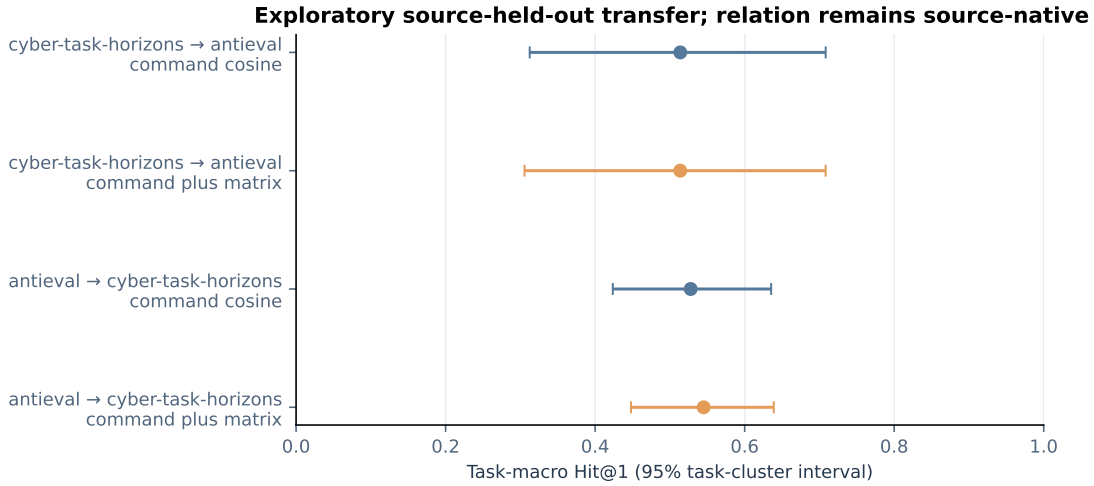


Figure 7: Exploratory transfer scores. Chance differs because candidate counts are three and five. The zero observed gain in CTH-to-Antieval is based on 12 test tasks and should not be read as proof of equivalence.

D Reproduction and artifacts

The run used the frozen protocol and fixed seed 20261102; task-cluster intervals use 2,000 bootstrap resamples. The final runner completed recognition, mixture, and both Agent-only retrieval directions. Early runner failures were cohort/fold bookkeeping errors and are retained in the run ledger; they are not experimental outcomes. Reproduce from the repository root:

```
.venv/bin/python research/experiments/overnight_fingerprint_v1/run.py all
.venv/bin/python research/experiments/overnight_fingerprint_v1/build_figures.py
```

Full predictions, fold assignments, eligibility lists, metrics, and hashes are in `research/results/overnight_fingerprint_v1/`. The associated research and source notes are `site/local-mode-detector.html#results` and `site/related-work.html#round-results`.

References

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